The term "Algebra" has evolved as a misspelling of the word 'al-jabr' from one of the important work titled Al-Kitāb al-mukhtaṣar fī hisāb al-jabr wa'l-muqābala ("The Compendious Book on Calculation by Completion and Balancing") written by Persian Mathematician Al-Khwarizmi of 9th Century AD(CE) Since Al-Khwarizmi's Al-Jabr book provided the most appropriate methods of solving equations, he is hailed as "Father of Algebra".

In the earlier classes, we had studied several important concepts in Algebra. In this class, we will continue our journey to understand other important concepts which will be of much help in solving problems of greater scope. Real understanding of these ideas will benefit much in learning higher mathematics in future classes.

Simultaneous Linear Equations in Two Variables

Let us recall solving a pair of linear equations in two variables.

Definition

Linear Equation in two variables

Any first degree equation containing two variables x and y is called a linear equation in two variables. The general form of linear equation in two variables x and y is $ax+by+c = 0$, where atleast one of $a,\ b$ is non-zero and $a,\ b,\ c$ are real numbers.

Note that linear equations are first degree equations in the given variables.

Note

- $xy - 7 = 3$ is not a linear equation in two variables since the term xy is of degree 2.
- A linear equation in two variables represent a straight line in xy plane.

Example 3.1 The father's age is six times his son's age. Six years hence the age of father will be four times his son's age. Find the present ages (in years) of the son and father.

Solution Let the present age of father be x years and

the present age of son be y years

Given, $x = 6y$ … (1)

$x + 6 = 4(y + 6)$ … (2)

Substituting (1) in (2), $6y + 6 = 4\left(y + 6\right)$

$6y + 6 = 4y + 24 \quad \Rightarrow \quad y = 9$

Therefore, son's age $=$ 9 years and father's age $=$ 54 years.

Example 3.2 Solve $2x - 3y = 6$, $x + y = 1$

Solution $2x - 3y = 6$ … (1)

$x + y = 1$ … (2)

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